

Weekly Report

How Detection Changed and How Earlier Results Leaked

Yifei Xu

New York University

October 2, 2026

This Week's Changes

- **Leak found:** in the earlier runs, *which data existed* told the agent the fault family. Detection itself was never tested.
- **New detection:** balanced SCADA, WLS and a balanced hypothesis screen. Auxiliary data opens only on a balanced suspicion, and every root carries it.
- **Expert under the new contract:** **159/160** development roots, no false commit.
- **LLM pipeline:** rerun from scratch with the new teacher. First honest student score: **R1 158/160** (BC0 152, expert 160). Round 2 is running.
- **Screen cost:** about 1 s per alarm. A gradient-boosted model on the WLS alone matches the request decision but not the HIF line; cheaper physics tests come first.

Earlier Scores and What They Assumed

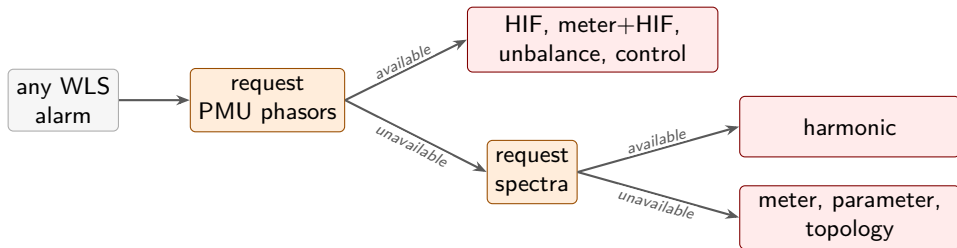
Run	Evidence contract	Expert	Student	What the agent was given
Sep 21–22	auxiliary-assisted	158/160	R2 157/160	HIF relay flag at reset; all auxiliary data always available
Sep 24–27	WLS-gated	158/160	R2 157/160	no flag, but phasors and spectra existed only on their own family's roots
Sep 22 check	SCADA-only	$\leq 104/160$		– no auxiliary data at all

- Neither 157/160 measures detection from balanced evidence.
- SCADA-only is the opposite extreme: the 56 HIF, meter+HIF, unbalance and harmonic roots are unreachable.

Source: S1, S6, S8. Sep 21–22 scores use the revised runtime and audit of Sep 22.

Leak 1: The Request Answered the Family Question

WLS-gated contract, September 23–27



- Routing rule: after *any* WLS alarm, request phasors; if unavailable, request spectra.
- Each stream existed only where the ground truth needed it, so one or two answers classified the root.
- HIF, unbalance and harmonic were **16/16 for expert, BC0, R1 and R2**, with the family known before any balanced evidence was weighed.

Source: S1 (“Availability-based cues”), S6.

Other Leaks: Family Cues in the Data and Tools

Cue	What it gave away	Status
Bus-9 capacitor counted twice in OpenDSS exports	every legacy HIF and unbalance row alarmed through bus-9 Q (median $J = 246$, limit 130); without it only 75/220 unbalance rows alarm	fixed Sep 19–21
Generator kW write reset its reactive limits	bus-8 $Q_{inj} \approx 0$ on every HIF row, 0.14–0.24 pu elsewhere	fixed Sep 23
OpenDSS families at default dispatch	bus-3 P and bus-8 V separated them from the OPF families	fixed Sep 23
HIF replay without voltage regulation	GNN pure-HIF recall 64.1% $\rightarrow$ 46.9% once regulation is emulated	GNN blocked in strict profiles
HIF estimators replayed the simulator operating point	truth inside the diagnosis; scan windows only on HIF roots	removed Sep 27

The capacitor made the WLS alarm itself an artifact. The bus-8 and dispatch cues let a learned model read the family from SCADA values.

Source: S1, S9, S10, S11.

How the Detection Contract Changed

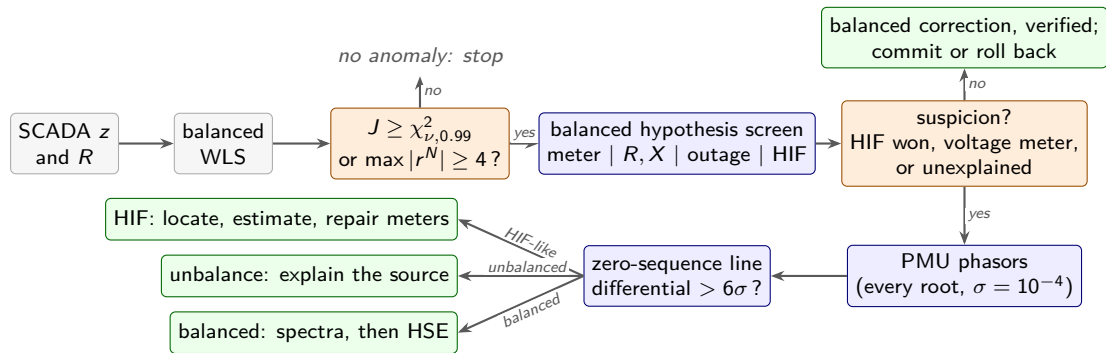
Date	Contract	Detection	Auxiliary data opens on	Carried by
to Sep 21	auxiliary-assisted	HIF relay flag; WLS	always	own family
Sep 22	SCADA-only	WLS alarm	never	–
Sep 23	WLS-gated	WLS alarm	any alarm	own family (leak)
Sep 27	suspicion-gated	WLS + HIF screen	HIF suspicion	every root
Sep 30	suspicion-gated v2	WLS + hypothesis screen	HIF, voltage-meter or unexplained; spectra after balanced phasors	every root
Oct 1	+ ledger and ranker	same	same; the ranker may defer one request	every root

Alarm test throughout:

$$J = e^T R^{-1} e \geq \chi_{\nu, 0.99}^2 \quad \text{or} \quad \max_i |r_i^N| \geq 4.$$

Source: S1, S2, S3.

The New Detection Pipeline



- Everything before the phasor box uses only SCADA, its covariance and the balanced model.
- A balanced correction that verification rejects also opens the phasors.

Source: S2, S3, S11.

Balanced Hypothesis Screen

After a WLS alarm, refit the operator's balanced model under competing single causes:

Class	Refit	Free parameters
Meter	one of the six largest residual channels set aside	its bias
Parameter	one branch's R , X , or both, inside a plausible box	1 or 2
Topology	one branch switched out, never islanding a bus	–
HIF	line split at α , shunt $G \geq 0$ at a new zero-injection bus	G

$$s_c = J_0 - J_c - (13.5 n_{\text{cont}} + 0.25 \ln n_{\text{cand}})$$

- Apply the winner and repeat, up to three comparisons, until the alarm clears or HIF wins.
- Outputs: accepted sequence (e.g. parameter>meter), HIF line, voltage channels set aside, *unexplained*.

Source: S3, S11 (hif_screen.py; penalties tuned on the Sep 26 design half).

What Opens Auxiliary Data Now

Request	Admitted only when
PMU phasors, phasor test	HIF won; a phase-A voltage channel was set aside; the alarm stayed unexplained; or verification rejected the screen's explanation
HIF estimator	screen HIF, or a zero-sequence differential above 6σ
Spectra, HSE	phasors on this state came back balanced
Voltage-meter edit	waits for phasors while a phasor suspicion stands

- Every root carries true-state phasors and noise-only spectra, so **no request answers by availability**.
- The phasor test uses each line's terminal differential: a fault to ground leaves a zero-sequence part; a balanced model error leaves none.

Source: S2, S3, S11 (suspicion_gated.py, evidence_profile.py).

Same Root Types, Old and New Traces

Root	WLS-gated (Sep 23)	Suspicion-gated (now)
Unbalance	alarm → phasors available → phasor test → explain	alarm → screen picks a voltage meter → phasors → phasor test finds the unbalance → explain
Harmonic	alarm → phasors unavailable → spectra available → HSE	alarm → screen: unexplained → phasors come back balanced → spectra → HSE
Bad power meter	alarm → phasors unavailable → spectra unavailable → meter route	alarm → screen picks the meter → meter route, no request

The old answers did the classification. The new answers are measurements the agent has to interpret, and a request needs a balanced reason.

Source: S1, S3 (expert traces under both contracts).

What Balanced Evidence Can Separate

Screen on the step-1 dataset: 2,148 alarmed fault roots and 69 alarmed healthy windows

Family	Alarmed	HIF flag	Voltage-meter pick	Unexplained	Phasor rule fires
HIF	131	130	0	0	130
Meter + HIF	131	128	0	0	128
Unbalance	162	0	160	77	161
Harmonic	250	0	248	208	249
Single meter	250	0	33	0	33
Multiple meters	224	0	28	113	126
Topology	250	0	11	55	55
Parameter, mixed balanced	750	0	2	0	2
Healthy windows	69	1	10	2	13

The HIF flag fires on 258 of 262 HIF roots and on no other fault root. Unbalance and harmonic look like a bad voltage meter, so phasors decide. The rule fires on 216 of the 1,474 balanced-family roots.

Source: S3 (screen v3, as shipped).

Expert Results Under Each Contract

Rule-based expert, 160 development roots, deployment-mode verifier, 40 actions

Contract	Note	Expert
WLS-gated (Sep 24)	family known from availability	158/160
SCADA-only	waveform families unreachable	$\leq 104/160$
HIF-only gate (Sep 27)	unbalance and harmonic uncredited	$\leq 128/160$
Screen v2 + rule v3	baseline expert	159/160
Screen v2 + rule v3	hypothesis-ledger expert	159/160
Same, + learned ranker	HPC teacher, cell's own draw	160/160

- Fresh check: **48/48** roots over all 12 families; unbalance in 4 steps, harmonic in 6.
- Ledger: 8.1 vs 8.8 steps per episode, 8 vs 15 rejected candidates; no false commit in either.
- The one miss is an adjacent-line parameter root that both rankings get wrong.

Source: S2, S3, S4, S6. The Sep 24 row is that cell's own draw; the rule-v3 rows share a fresh draw with the same family plan. Ceilings count unreachable families.

Fewer Unneeded Acquisitions: Learned Ordering

Gradient-boosted model on 73 policy-visible features; test split, 463 alarmed roots

	Physics rule	Learned score
Recall, roots that need auxiliary data	98.5%	99.2%
Acquisitions on roots that need none	12.0%	2.1%
Same-sign flow-meter pairs flagged as HIF	37.5%	0%

- The physics rule stays the gate. The score only lets the expert try its leading balanced hypothesis once before an admitted request.
- Paired on 208 roots: **63 vs 72** phasor requests at equal success; no root that needs phasors was deferred.
- IEEE-57 operating points do not transfer; the model refuses another network.

Source: S5.

What the Screen Costs

IEEE-14; opening WLS of the 144 alarmed development roots; one core

Class	Candidates per round	Refits
Meter	six largest residuals	6
Parameter	20 branches $\times$ R , X , both	60
Outage	branches that island no bus	19
HIF	16 lines $\times$ 9 positions	144
One round		229

	Median	Max
WLS solve alone	0.05 s	0.43 s
Screen	1.1 s	3.0 s
Refits per screen	448	692

- Every refit is a full constrained WLS of the modified model, started from the alarmed solution; cached per state.
- 64 screens stop after one round, 37 after two, 43 after three; with two meters set aside a round costs 85 refits.
- Collection takes about 33 s per root, so on IEEE-14 the screen is a few seconds; refits grow with the branch count.

Source: S12 (commit a76f704), S11.

Can a Gradient-Boosted Model Replace the Screen?

Step-4 dataset and splits; 463 alarmed test roots, 130 needing phasors or spectra, 50 with an HIF

Decision from	Request gate		HIF flag	
	Recall	Unneeded requests	Recall	False flags
Physics rule and screen	98.5%	12.0%	98%	1.5%
Gradient-boosted, alarmed WLS only	98.5%	10.5%	78%	0.5%
Gradient-boosted, plus screen outputs	96.9%	1.2%	98%	1.7%

- With no refit, a model on residuals, multipliers and flow pairs matches the rule on *whether* to request phasors.
- It misses the HIF line: at the screen's false-flag level it flags 39 of 50 HIF roots, against 49.
- The screen adds the HIF line and the ordered hypotheses; a model on its outputs cuts unneeded requests to 1.2%.
- A learned-only gate needs retraining per network and can pick up simulator artifacts that name the family.

Source: S12, S5. 45 single-solve features vs. the study's 94; thresholds from the calibration split. Slide 13: deployed 73-feature model.

Suggested Fixes

Fix	How	Saves	Status
1. Linearized tests, refit a shortlist	Residuals score meters and Lagrange multipliers score branches, both from the alarmed WLS; an injection test scores HIF positions; refit the top one or two per class	most of the 229 refits per round, about tenfold	measure agreement with the full screen
2. Two-stage gate	Screen-free model on every alarm; full screen only when it is uncertain or predicts phasors	the screen on confident balanced roots	to evaluate
3. Parallel refits	Candidates are independent; about 16 cores per GPU on the cluster	wall time	engineering
4. Spectra before a voltage-meter edit	When phasors come back balanced on a voltage-meter suspicion, request spectra first	16 of the 26 extra actions below	next teacher

- Expert today: 1,270 actions on 160 development roots against a shortest 1,244; 4 rollbacks, all on two harmonic roots.
- Order: fix 1 first, as it keeps the physics decisions and works on any network; fix 2 if larger networks stay slow.

Source: S12 (teacher's own paths at commit a76f704).

Remaining Cues and Limits

- **Availability cues on balanced families remain:** repeated parameter scans exist only on parameter roots, breaker telemetry only on topology roots.
- **No balanced class for unbalance or harmonic:** both read as a bad voltage meter; only phasors and spectra separate them.
- **Opposite-sign flow-meter pairs:** the screen prefers a branch hypothesis (19/24 vs 21/24 for the static order).
- **Detection limit:** admitted HIFs are mostly 100–200 Ω ; above 200 Ω the WLS alarm, not the screen, is the bottleneck.
- **IEEE 57:** the HIF class wins on only 16 of 112 alarmed HIF roots.
- **Model match:** the shunt hypothesis matches the simulator's resistive HIF; arcing faults are untested.

Source: S1, S3, S5, S8.

Current HPC Run: LLM Under the New Contract

Gemma 4 12B, IEEE-14, suspicion-gated contract, ledger-plus-ranker teacher

Stage	Result
D0 expert aggregate	514 roots, 4,421 rows (3,350 train)
BC0	838 steps; best validation loss 0.0019
Round-1 collection	122 roots; label yield 0.88; 1,112 mixture rows
Round-1 evaluation	expert 160/160 , BC0 152/160 , R1 158/160 ; every success terminal
Round 2	collection running

- R1 vs BC0: false-commit episodes 1 vs 2, invalid-action episodes 5 vs 13, loop episodes 2 vs 9.
- R1 misses one parameter root (BC0 misses it too) and one topology root.

Source: S7 (out/r1/round_summary.json, evaluation files). Status at 14:40 UTC, October 2.

Next Experiments

- 1 **Finish round 2.** Compare R2 with R1 and the ledger teacher on the same 160 roots, by success basis.
- 2 **Cheaper screen.** Linearized tests with a refit shortlist; measure winner agreement, then try the two-stage gate.
- 3 **Next teacher.** Spectra before a voltage-meter edit when the phasors come back balanced.
- 4 **Remove the last availability cues.** Parameter scans and breaker telemetry on every root, or gated on a suspicion.
- 5 **Close the flow-pair gap.** Add a two-meter hypothesis to the screen.
- 6 **IEEE 57.** Candidate lines for the screen; retrain and recalibrate the ranker.

Appendix: The WLS-Gated Run by Family

Family	N	Expert	BC0	R1	R2
Measurement	16	16	16	15	16
Multiple measurements	12	12	12	12	12
Parameter	16	14	14	14	14
Measurement + parameter	16	16	14	16	16
Topology	16	16	16	16	16
Measurement + topology	12	12	12	12	12
HIF	16	16	16	16	16
Measurement + HIF	8	8	8	8	7
Unbalance	16	16	16	16	16
Harmonic	16	16	16	16	16
No-error controls	8	8	8	8	8
Healthy telemetry controls	8	8	8	8	8
Total	160	158	156	157	157

Source: S6 (out/pipeline_summary.json), Sep 24–27 cell, 160 development roots. Shaded rows: the request outcome named the family.

Appendix: Sources

- S1 docs/wls_gated_evidence_20260923.md: WLS-gated contract, root contents, known family cues.
- S2 docs/hypothesis_ranking_plan_20260930.md: contract review, decisions C1, C2, D3, C5, step outcomes.
- S3 docs/hypothesis_ranking_step1_20260930.md, step2_20260930.md: screen results and acquisition rule.
- S4 docs/hypothesis_ranking_step3_20261001.md: ledger expert, paired 160-root evaluation.
- S5 docs/hypothesis_ranking_step4_20261001.md, step5_20261001.md: learned ranker and paired arms.
- S6 Cluster cell research_full_pipeline_20260924_wls_gated, out/pipeline_summary.json.
- S7 Cluster cell research_full_pipeline_20261001_ranked: d0.done, bc0.done, round-1 receipts.
- S8 docs/slides/weekly_update_20260923_sources.md: Sep 21–22 run and revised audit.
- S9 docs/ieee14_hif_legacy_reconfiguration_20260919.md: capacitor and reactive-limit artifacts.
- S10 research/gnn_screen/results/practical_v2_20260916/metrics.json; GNN review of Sep 22.
- S11 psse_env/providers/hif_screen.py, suspicion_gated.py, psse_env/evidence_profile.py.
- S12 output/hypothesis_ranking_20260930/screen_cost/: screen timing, screen-free models, teacher paths (Oct 2).